Primitive residues in rank two

Human Compression project

Working proof, 14 October 2026

Let G be finite. We use right cosets, with incidence given by nonempty intersection. Firmness means that every panel lies in at least two chambers. For a rank-two geometry, residual weak primitivity requires a primitive G-action on at least one type and primitive actions of both vertex stabilizers on their respective neighbours. Thus it includes conditions on the rank-one residues. No double transitivity is assumed.

Theorem 1. *All connected, firm, flag-transitive, residually weakly primitive rank-two geometries for G satisfying the intersection property are obtained by the following construction.*

1. *Choose one representative A of each conjugacy class of maximal proper subgroups of G.*
2. *Choose one representative H of each A-conjugacy class of maximal proper subgroups of A.*
3. *Take every actual subgroup B with $H < B < G$, retaining it precisely when $A \cap B = H$ and H is maximal in B.*
4. *Retain the pairs passing Lemma 2. Identify pairs under automorphisms of G and interchange of their entries.*

The final identification classifies the specified G-actions. Distinct resulting classes may require a further comparison if one wishes to forget the specified action and classify abstract incidence geometries.

Proof. Choose an incident pair in such a geometry. Its stabilizers A, B identify the geometry with the incidence structure of their cosets, by flag transitivity. Interchange the types so that A is maximal in G, and conjugate to the chosen representative. The common stabilizer $H = A \cap B$ is maximal in both A and B: in a transitive permutation action, primitivity is equivalent to maximality of a point stabilizer. Conjugating within A now puts H in the second list. The resulting B is one of the actual overgroups required in the third step. This proves completeness before the incidence test and equivalence comparison.

Conversely, the two proper containments $H < A, B$ give firmness. Since $H < B$ and $A \cap B = H$, the subgroup B is not contained in A. Maximality of A therefore gives $\langle A, B \rangle = G$, which proves connectivity. Every incident pair of cosets contains a common element, so it is a translate of (A, B); hence the action is flag-transitive. The three maximality conditions give precisely the primitive actions required above. These are all the nonempty residues that need checking in rank two. Lemma 2 supplies the intersection property. An automorphism carrying one pair to another, possibly interchanging its entries, induces the corresponding incidence isomorphism. □

Lemma 2. *For a pair obtained in the third step, let T_A, T_B be right transversals of H in A, B. For each $a \in T_A \setminus H$, put*

$$c(a) = \big|\{b \in T_B : \text{ some } d \in T_B \text{ has } b(da)^{-1} \in A\}\big|.$$

The intersection property holds exactly when all values $c(a)$ equal 1, or when $[B : H] = [G : A]$.

Proof. The neighbours of B are Ab for $b \in T_B$, while those of Ba are Ada for $d \in T_B$. Both lists have distinct entries because $A \cap B = H$. Equality of these two right cosets is exactly the displayed membership test, so $c(a)$ counts their common neighbours. The cosets Ba with $a \in T_A$ exhaust the neighbours of A. The group A is transitive on these neighbours; consequently every pair of distinct neighbours of A can be moved to (B, Ba) for one of the tested cosets. Transitivity on vertices of the first type now shows that the condition $c(a) = 1$ for all a is equivalent to having no quadrangle anywhere in the incidence graph. This is the partial linear space condition.

The other allowed possibility is a generalized digon, namely a complete bipartite geometry. A vertex of the second type has $[B : H]$ neighbours among $[G : A]$ vertices of the first type. By transitivity, completeness is therefore equivalent to equality of these numbers. The two alternatives are exactly the required intersection property. □

Application to $\mathrm{PSL}(3, 4)$

The computation uses the degree-21 permutation representation supplied by GAP. It acquires the complete maximal-subgroup algorithms and data for G, its maximal subgroups, and the intermediate groups encountered during the overgroup search. These subgroup classifications are retained inputs; the argument above does not reprove them. In particular, the routine for intermediate subgroups must return actual subgroups, rather than only conjugacy representatives.

There are nine conjugacy classes of maximal subgroups. Their local lists give 42 pairs (A, H). The third step yields 60 pairs (A, B), of which 31 pass the incidence test. They give the following seven classes. The source labels refer to the complete Halle preprint of Gottschalk and Leemans, *Geometries for the group PSL(3,4)*, Report 99–30 (1999), pp. 7–8; its later journal edition has not been collated here.

Source	$\lvert A\rvert$	$\lvert B\rvert$	$\lvert H\rvert$	$[A : H]$	$[B : H]$	$(2T)_1$
2.1	960	960	192	5	5	yes
2.2	360	96	24	15	4	no
2.3	360	72	36	10	2	yes
2.4	168	168	21	8	8	yes
2.5	168	96	24	7	4	yes
2.6	72	72	8	9	9	yes
2.7	72	16	8	9	2	yes

Both local actions are doubly transitive in six of the seven classes. Class 2.2 has a primitive action of degree 15 whose point stabilizer has order 24; it cannot be doubly transitive, since 14 does not divide 24. Its other local action, of degree 4, is doubly transitive. These are outputs of the computation: the search includes every primitive local action and rejects pairs only by the stated tests. The seven unordered pairs of parabolic orders are distinct. Since the two type sizes are $[G : A]$ and $[G : B]$, the seven representatives are also inequivalent as abstract incidence geometries, even allowing interchange of types. Explicit automorphisms identifying the other surviving pairs are supplied separately as images of generators.

The coset correspondence and the stabilizer criterion for primitivity are classical. The present argument organizes these inputs into the same finite construction used for doubly transitive residues, with the incidence test extended to all local pairs. No claim of historical novelty for this enumeration principle is made.